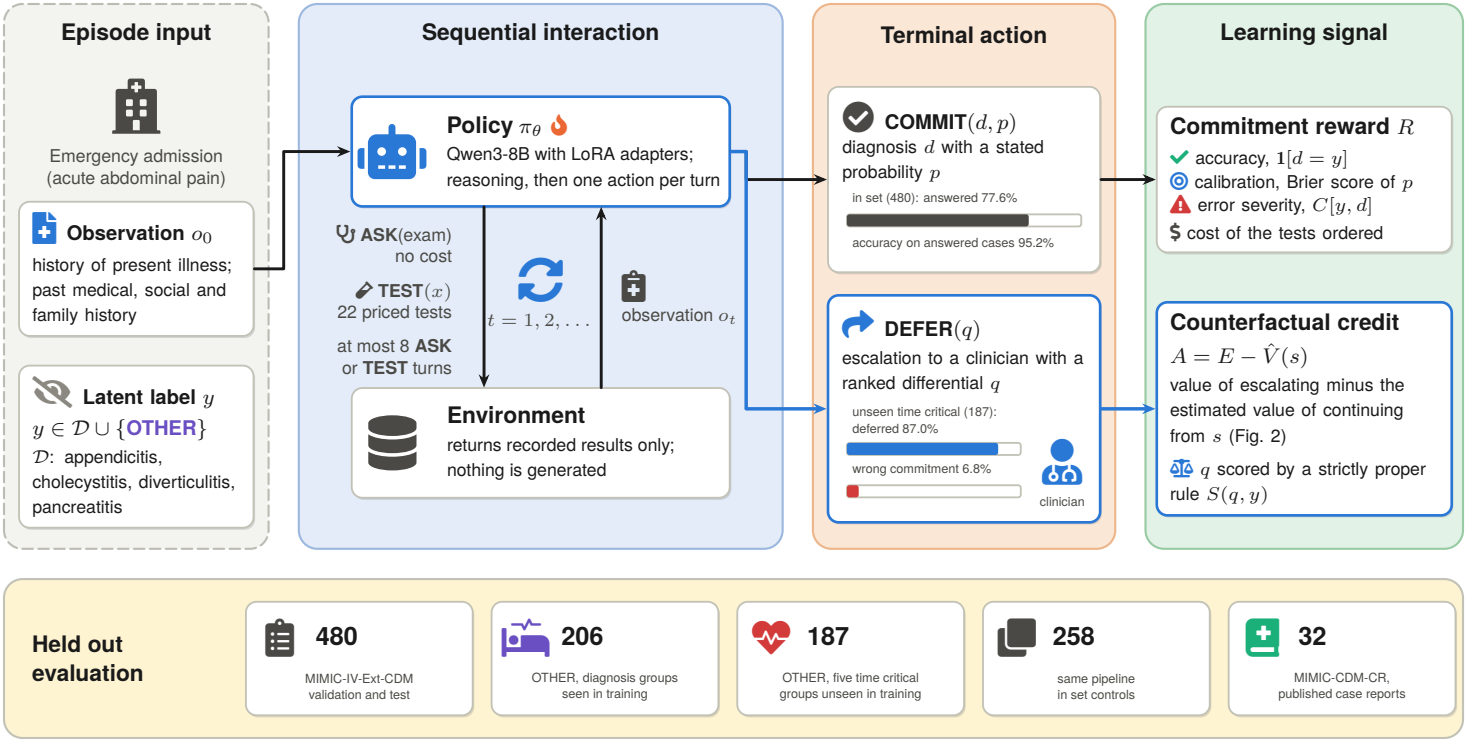
Episode input
Emergency admission
(acute abdominal pain)
Observation $o_0$
history of present illness; past medical, social and family history
Latent label $y$
$y \in \mathcal{D} \cup \{\text{OTHER}\}$
$\mathcal{D}$: appendicitis, cholecystitis, diverticulitis, pancreatitis
Sequential interaction
Policy $\pi_\theta$
Qwen3-8B with LoRA adapters; reasoning, then one action per turn
ASK(exam) no cost
TEST($x$) 22 priced tests
at most 8 ASK or TEST turns
$t = 1, 2, \ldots$
observation $o_t$
Environment
returns recorded results only; nothing is generated
Terminal action
COMMIT$(d, p)$
diagnosis $d$ with a stated probability $p$
in set (480): answered 77.6%
accuracy on answered cases 95.2%
DEFER$(q)$
escalation to a clinician with a ranked differential $q$
unseen time critical (187): deferred 87.0%
wrong commitment 6.8%
clinician
Learning signal
Commitment reward $R$
accuracy, $\mathbf{1}[d = y]$
calibration, Brier score of $p$
error severity, $C[y, d]$
$ cost of the tests ordered
Counterfactual credit
$A = E - \hat{V}(s)$
value of escalating minus the estimated value of continuing from $s$ (Fig. 2)
$q$ scored by a strictly proper rule $S(q, y)$
Held out evaluation
480
MIMIC-IV-Ext-CDM validation and test
206
OTHER, diagnosis groups seen in training
187
OTHER, five time critical groups unseen in training
258
same pipeline in set controls
32
MIMIC-CDM-CR, published case reports